JAN-S-350

31 MAY 1946

JOINT ARMY-NAVY SPECIFICATION

SPRINGS, SLIDE, 4.2-INCH CHEMICAL MORTAR BARREL, C3 AND C4

Army Number
96-81-226A

Navy Number
1S3

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specification:

U. S. Army
96-81-226
14 Dec. 1944

Navy Department
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A. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

A-1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN-P-106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
JAN-P-116—Packaging and Packing for Overseas Shipment—Preservation, Methods of.
JAN-P-117—Packaging and Packing for Overseas Shipment—Bags, Interior Packaging.
JAN-P-140—Packaging and Packing for Overseas Shipment — Adhesives, Water-Resistant, Case Liner.

U. S. ARMY SPECIFICATIONS

19-8 —Liner, Shipping Container, Waterproof Barrier.
19-10—Filler, Interior Packing, Fiberboard, Corrugated, Single and Double Faced, for General Packing Purposes.
57-107—Steel, Bar, Carbon and Alloy.
100-2—Standard Specification for Marking Shipments by Contractors.[1]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

A-2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[2]

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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A–3. *Drawings.*—The following drawings, of the date indicated, form a part of this specification.

CHEMICAL WARFARE SERVICE DRAWINGS.

Number	Date of approved drawing or revision	Title
E8–2–146	26 June 1944	Spring, Front, C3
E8–2–147	26 June 1944	Spring, Rear, C4

B. TYPE AND CLASSES

B–1. The steel compression springs shall be of one type and of the following classes as defined by this specification (See pars. H–1 and H–2):

Class a—Front spring, C3.
Class b—Rear spring, C4.

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—The steel used to manufacture the springs shall be the WD 9260 composition prescribed in U. S. Army Specification 57–107, or SAE 9255. The steel stock shall be drawn in a form suitable for winding springs in conformance with the requirements of this specification.

C–2. *Workmanship.*—The workmanship shall be first class in every respect.

D. GENERAL REQUIREMENTS

D–1. *Manufacture.*—The springs shall be manufactured to meet the following requirements:

D–1a. *Winding.*—Each spring shall be wound right hand and the pitch of the coil shall be uniform.

D–1b. *Straightness.*—Each spring shall be sufficiently straight and uniform in diameter throughout its entire length to pass the test specified in paragraph F–4b.

D–1c. *Nonclosing load.*—Each spring shall support a dead load of at least 245 pounds without closing completely.

D–2. *Hardness.*—The steel in the finished springs shall have a Rockwell C scale hardness of 48 to 52.

E. DETAIL REQUIREMENTS

E–1. *Class a.*—Class a springs shall conform with the requirements of Chemical Warfare Service Drawing E8–2–146.

E–2. *Class b.*—Class b springs shall conform with the requirements of Chemical Warfare Service Drawing E8–2–147.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. The material and work shall be subject at all times to inspection at the place of manufacture and every facility shall be afforded authorized Government inspectors for the prosecution of their work.

F–2. *Size of lot.*—A lot shall consist of 5000 front and 5000 rear springs, unless otherwise specified in the contract or order.

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F–3. *Sampling.*—Samples for hardness determination shall be taken as follows: Not less than 1 spring shall be taken at random from those heat treated in each hour of production under constant conditions, but in no case shall the total number of samples taken exceed 1 percent of the entire production.

F–4. *Inspection and tests.*—All inspection and testing, other than visual examination, shall be done before the varnish is applied to the spring. Compliance with the requirements of this specification shall be determined as follows:

F–4a. *Winding.*—Each spring shall be examined visually for direction of wind and uniformity of coil pitch.

F–4b. *Straightness.*—Each spring shall be inspected for straightness and uniformity of diameter as follows: A straight tube as long or longer than the spring, having an inside diameter of 0.980 ± 0.003 inch, shall be set up vertically. The spring shall pass through the tube by its own weight. A round rod 0.510 ± 0.002 inch and as long or longer than the spring shall be set up vertically. The full length of the spring shall pass over the rod by its own weight. Each spring failing to pass through the tube or over the rod shall be rejected.

F–4c. *Design and dimensions.*—The springs shall be measured and inspected for compliance with the applicable drawing. (See pars. E–1 and E–2.)

F–4d. *Finish.*—The finished spring shall be examined visually for compliance with the finish requirement specified on the applicable drawing. (See pars. E–1 and E–2.)

F–4e. *Nonclosing load.*—Each spring shall be placed over a ½-inch rod, compressed under a dead load of 245 pounds, and examined visually to ascertain that the spring is not completely closed.

F–4f. *Hardness.*—The springs selected for hardness determination shall have a portion of 1 coil taken from each sample near the middle of the spring as a test specimen. The specimens shall have 2 opposite sides, ground until faces are parallel. The ground specimens shall be tested for hardness on a standard Rockwell Hardness Tester. All groups of springs represented by a specimen which is not within the hardness limits shall be rejected.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *For shipment to 4.2-inch chemical mortar barrel assembly plants.*—

G–1a. *Packaging.*—Two front springs and 2 rear springs (1 set) shall be packaged together in a grade A, class C greaseproof-waterproof envelope, conforming to the requirements of Joint-Army Navy Specification JAN–P–117, which shall be heat sealed.

G–1b. *Packing.*—Twenty-five packages (see par. G–1a) shall be packed in a style 2, 2½ or 4 nailed wood box conforming to the requirements of Joint Army-Navy Specification JAN–P–106, having approximate inside dimensions of 15 by 10¼ by 5¾ inches. The top, sides and bottom shall be made of ½-inch lumber, the end of ¾-inch lumber, and the cleats of ¾ by 2¼-inch lumber.

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The box shall be lined with double faced corrugated fiberboard filler conforming to the requirements of U. S. Army Specification 19–10. One filler shall be inserted between each layer of packages. The closed box shall be bound, not more than one week before shipment, with 2 bands of ⅝ by 0.020-inch nailless metal strapping.

G–2. *For shipment as spare parts.*—

G–2a. *Packaging.*—Two front or 2 rear springs shall be packaged together in accordance with method IA–1 of Joint Army-Navy Specification JAN–P–116.

G–2b. *Packing.*—Twenty-five packages of either 2 front or 2 rear springs each, shall be packed in a style 2, 2½ or 4 nailed wood box conforming to the requirements of Joint Army-Navy Specification JAN–P–106. The top, sides and bottom shall be made of ½-inch lumber, the ends of ¾-inch lumber, and the cleats of ¾ by 2¼-inch lumber. The box shall be lined with a waterproof paper liner conforming to the requirements of U. S. Army Specification 19–8 and 1 thickness of double-faced corrugated fiberboard filler, conforming to the requirements of U. S. Army Specification 19–10 shall be placed between the waterproof liner and all surfaces of the load. The liner shall be folded down on the contents of the box, and all contacting surfaces shall be completely sealed with water resistant adhesive conforming to the requirements of Joint Army-Navy Specification JAN–P–140. The closed box shall be bound, not more than one week before shipment, with 2 bands of ⅝ by 0.020-inch nailless metal strapping.

G–3. *Marking.*

G–3a. *Marking of individual packages.*—Each individual package shall be labeled as to contents, lot number, and manufacturer's identification mark.

G–3b. *Marking of shipping containers.*—In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy, in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. Requests, requisitions, schedules and contracts or orders should specify the following:

(a) The title of the specification, the number and date.
(b) That all springs are to be furnished in sets, each set consisting of 2 Class a and 2 Class b springs. (See par. B–1.)
(c) Whether to be packaged and packed for shipment to assembly plants or as spare parts (see section G).

H–2. One set of springs is used as the cushioning element in the shock absorber on a 4.2-inch chemical mortar barrel.

H–3. Copies of Chemical Corps drawings may be obtained from the New York Chemical Corps Procurement District Office or from the office of the Chief, Chemical Corps, Washington 25, D. C.

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H–4. Copies of Joint Army-Navy specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy and U. S. Army specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H-5. Copies of Joint Army-Navy specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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